RDD2022 + YOLO26 Experiment Database

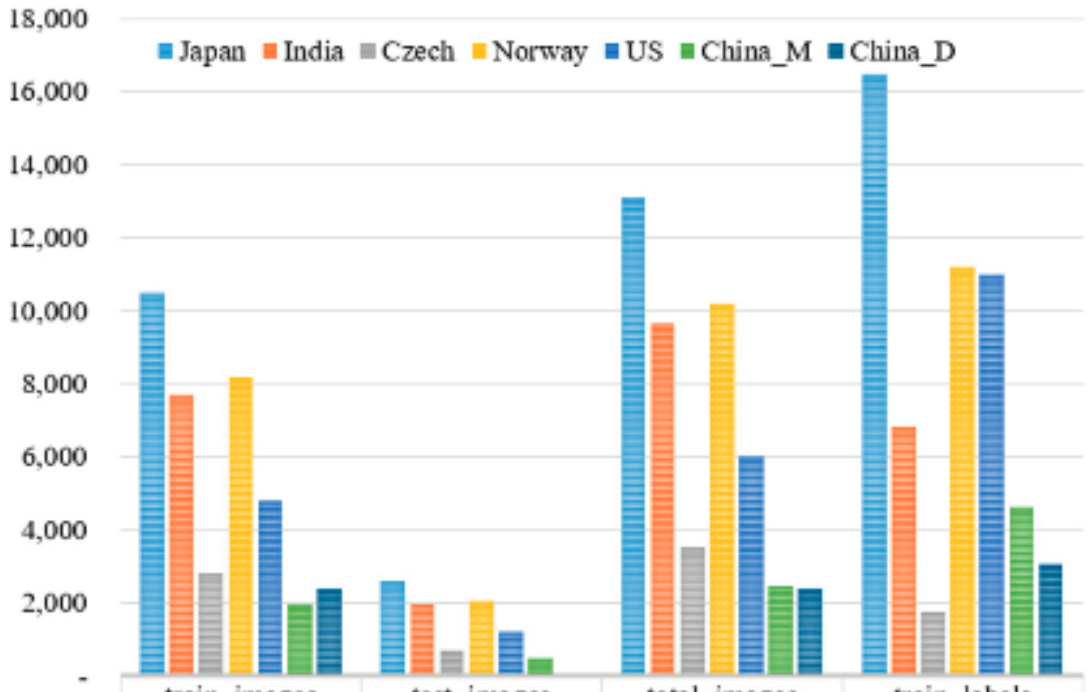
Clean Experiment Log

Researcher: Md. Tahmid Islam Tomal **Project:** Road Damage Detection with YOLO26

Last Updated: May 2, 2026

Quick Index

Database ID	Dataset	Main Purpose
DB-RDD22-CM-4C-V1	China MotorBike, 4 classes	YOLO26m baseline on China MotorBike.
DB-RDD22-CHINA-ALL-5C-V1	China All, 5 classes	YOLO26s baseline on China-only subset.
DB-RDD22-FULL-4C-V1	Full RDD2022, 4 classes	YOLO26s and YOLO26m paper-style comparison.
DB-RDD22-ALL-4C-V2	RDD2022 All, 4 classes	YOLO26s stricter reproduction with labeled test split.
DB-RDD22-JAPAN-4C-V2	Japan, 4 classes	YOLO26s stricter reproduction on Japan-only split.



	train_images	test_images	total_images	train_labels
Japan	10,506	2,627	13,133	16,470
India	7,706	1,959	9,665	6,831
Czech	2,829	709	3,538	1,745
Norway	8,161	2,040	10,201	11,229
US	4,805	1,200	6,005	11,014
China_M	1,977	500	2,477	4,650
China_D	2,401	-	2,401	3,068

Figure 1: Overall experiment database index and dataset record summary.

Dataset Records

RDD2022 China MotorBike Paper-Style YOLO26m Experiment `DB-RDD22-CM-4C-V1`

Short name: RDD2022-China-MotorBike-4C
Goal: Use YOLO26m as a baseline for a paper-style China MotorBike experiment.
Data: China MotorBike only. XML annotations were converted to YOLO format.
Split: 1450 train images, 242 validation images, 242 test images.
Classes: 4 classes: longitudinal crack, transverse crack, alligator crack, pothole.
Background images: Not used as a separate class. Empty-label images are treated as background by YOLO.
Cleaning: Invalid or empty labels were checked. Class mapping was limited to the 4 selected categories.

From the RDD2022 [1] dataset, we selected vehicle-captured road images from China to create an experimental dataset named RDD-China-Motorbike, comprising 2,477 images. Additionally, we extracted drone-captured Chinese road images from RDD2022 to form another dataset called RDD-China-Drone (2,401 images), which was used to validate the model's generalization capability. Since the official RDD2022 dataset does not provide annotations for its test set, we reorganized the publicly available portions of both datasets, splitting each into training, validation, and test sets at a ratio of 6:1:1.

Figure 2: China MotorBike dataset setup used for the YOLO26m paper-style baseline.

Table 1. Comparison Experiment

Model	Params	P	R	F1	mAP@0.5	mAP	FPS(bs=1)
Faster R-CNN [12]	28.3	57.3	63.6	60.3	80.9	42.3	41.1
CenterNet [3]	14.2	73.1	72.3	72.7	81.4	43.7	125.4
RetinaNet [8]	19.8	42.9	50.3	46.3	58.2	27.5	42.2
RTMDet_m [11]	24.7	82.7	80.6	81.6	83.1	45.5	69.0
YOLOv5_m [5]	25.2	86.5	80.5	83.4	83.5	51.0	138.7
YOLOv8_m [15]	25.9	90.9	86.8	85.4	91.6	58.9	122.0
YOLOv9_m [17]	19.1	90.7	85.4	88.0	89.7	56.8	109.9
YOLOv10_m [16]	15.7	82.9	83.5	83.2	88.6	57.6	121.5
YOLOv11_m [6]	19.1	87.7	87.4	87.5	90.3	57.6	128.2
Deformable DETR [20]	23.5	63.2	65.1	64.1	67.4	33.0	31.8
DAB-DETR [10]	30.9	49.0	43.5	46.1	49.1	24.9	23.5
DINO [18]	30.9	88.5	89.9	89.2	91.0	58.4	23.1
RT-DETR [19]	18.9	90.2	88.8	89.5	92.2	58.5	131.6
RDD-DETR	**13.5**	**92.3**	**90.5**	**91.4**	**93.2**	**63.7**	**138.9**

Figure 3: China MotorBike YOLO26m result comparison against the selected paper-style baseline.

Run: RUN-CM-001

Model	YOLO26m
Core setup	image size 640, batch 32, 200 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, paper-style/default YOLO augmentation
Result	Test mAP50 0.902, mAP50-95 0.599, precision 0.899, recall 0.904

Model	YOLO11m
Core setup	image size 640, batch 32, 200 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, paper-style/default YOLO augmentation
Result	Test mAP50 0.911, mAP50-95 0.605, precision 0.900, recall 0.888

RDD2022 China All Paper-Style YOLO26s Experiment `DB-RDD22-CHINA-ALL-5C-V1`

Short name: RDD2022-China-All-5C
Goal: Compare YOLO26s against YOLO/RDD paper baselines on a China-only subset.
Data: China MotorBike and China Drone images.
Source: `/content/drive/MyDrive/Research/Dataset/China_Subset.zip`. Extracted to `/content/China_Subset`.
Split: 3502 train images, 438 validation images, 438 test images.
Classes: 5 classes: longitudinal crack, transverse crack, alligator crack, pothole, repair.
Background images: 5 in train, 0 in validation, 0 in test. Not used as a separate class.
Cleaning: Already in YOLO format. Ultralytics scan reported 0 corrupt images. `nc=5`.

Dataset

In the experiment, we chose two parts of RDD2022 named RDD-China, a large open-source dataset containing road surface pictures of many countries, as our experimental dataset. The dataset contains a total of 4,378 pictures, of which 2,401 are taken by drones and 1,977 are taken by on-board cameras. As shown in Fig. 7, there are five types of road damage in the picture: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), potholes (D40), and road repairs. These images are divided into train_set, valid_set, and test_set in an 8:1:1 ratio.

Figure 4: China All 5-class dataset setup for the YOLO26s paper-style experiment.

Table 3 The performance of model with different size.

Models	mAP@0.5/%	Para/10^6	GFLOPs	F1-score/%	FPS
YOLOv8n	84.2	3.0	8.1	80.3	243
YOLOv8s	85.1	11.1	28.4	80.8	164
YOLOv8m	85.7	25.8	78.7	81.2	102
OBC-YOLOv8n	86	3.2	8.0	81.9	222
OBC-YOLOv8s	86.2	12.1	28.6	81.8	151
OBC-YOLOv8m	86.4	27.1	78.2	82.1	85

Figure 5: China All 5-class YOLO26s validation and test comparison.

Run: RUN-CHINA-ALL-001

Model	YOLO26s
Core setup	image size 640, batch 32, 300 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 16, NVIDIA L4
Augmentation	HSV enabled, translate 0.1, scale 0.5, fliplr 0.5, mosaic 1.0; degrees, shear, perspective, flipud, mixup, and copy-paste disabled
Validation	mAP50 0.883, mAP50-95 0.597, precision 0.864, recall 0.837
Test	mAP50 0.848, mAP50-95 0.569, precision 0.850, recall 0.811

Model	YOLO8m
Core setup	image size 640, batch 32, 300 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 16, NVIDIA L4
Augmentation	HSV enabled, translate 0.1, scale 0.5, fliplr 0.5, mosaic 1.0; degrees, shear, perspective, flipud, mixup, and copy-paste disabled
Test	mAP50 0.839, mAP50-95 0.552, precision 0.846, recall 0.811

Model	YOLO8s
Core setup	image size 640, batch 16, 300 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 16, NVIDIA L4
Augmentation	HSV enabled, translate 0.1, scale 0.5, fliplr 0.5, mosaic 1.0; degrees, shear, perspective, flipud, mixup, and copy-paste disabled
Test	mAP50 0.856, mAP50-95 0.545, precision 0.867, recall 0.813

Model	YOLO26m
Core setup	image size 640, batch 16, 300 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 16, NVIDIA L4
Augmentation	HSV enabled, translate 0.1, scale 0.5, fliplr 0.5, mosaic 1.0; degrees, shear, perspective, flipud, mixup, and copy-paste disabled
Test	mAP50 0.850, mAP50-95 0.567, precision 0.877, recall 0.804

RDD2022 All 4-Class Test Split YOLO26s Experiment `DB-RDD22-ALL-4C-V2`

Short name: RDD2022-ALL-4C-TestSplit
Goal: Evaluate YOLO26s on a labeled test split under stricter reproduction settings.
Data: Full prepared RDD2022_ALL dataset. Prediction outputs include China MotorBike images.
Source: `/content/drive/MyDrive/Research/Dataset/RDD2022_ALL.zip`. Extracted to `/content/RDD2022_ALL`.
Split: 15294 train images, 4370 validation images, 2184 test images. All splits had matching label files.
Classes: 4 classes: longitudinal crack, transverse crack, alligator crack, pothole.
Background images: 0 in train, validation, and test.
Cleaning: YOLO format. `nc=4`. Class IDs found: 0, 1, 2, 3. Test scan found 0 corrupt files.

Table 5. Comparison among different object detectors.

Model	F1 Score (%)	mAP@0.5 (%)	No. of Params. (M)	GFlops	FPS
YOLOv8l	69.06	71.50	43.70	165.2	23
YOLOv8m	68.63	70.79	25.80	78.70	32
YOLOv5l	68.78	70.90	46.50	109.5	26
YOLOv5m	66.55	69.23	21.20	49.00	36
YOLOv7-tiny	63.68	65.67	6.27	13.80	65
YOLOv5s	65.22	65.91	7.01	16.00	63
YOLOv8s	66.46	67.70	11.21	28.40	53
Ours Improved YOLOv5s	66.51	67.81	8.11	16.30	58

Figure 6: RDD2022 All 4-class labeled test-split setup and YOLO26s test comparison.

Run: RUN-ALL-4C-001

Model	YOLO26s
Core setup	image size 640, batch 32, 100 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, label smoothing 0.1, cache disk, workers 16, AMP true
Augmentation	Disabled: HSV, translate, scale, fliplr, mosaic, degrees, shear, perspective, flipud, mixup, copy-paste
Test	mAP50 0.425, mAP50-95 0.191, precision 0.555, recall 0.428
Test size	2184 images, 5172 instances
Per-class mAP50-95	longitudinal 0.212, transverse 0.171, alligator 0.264, pothole 0.117

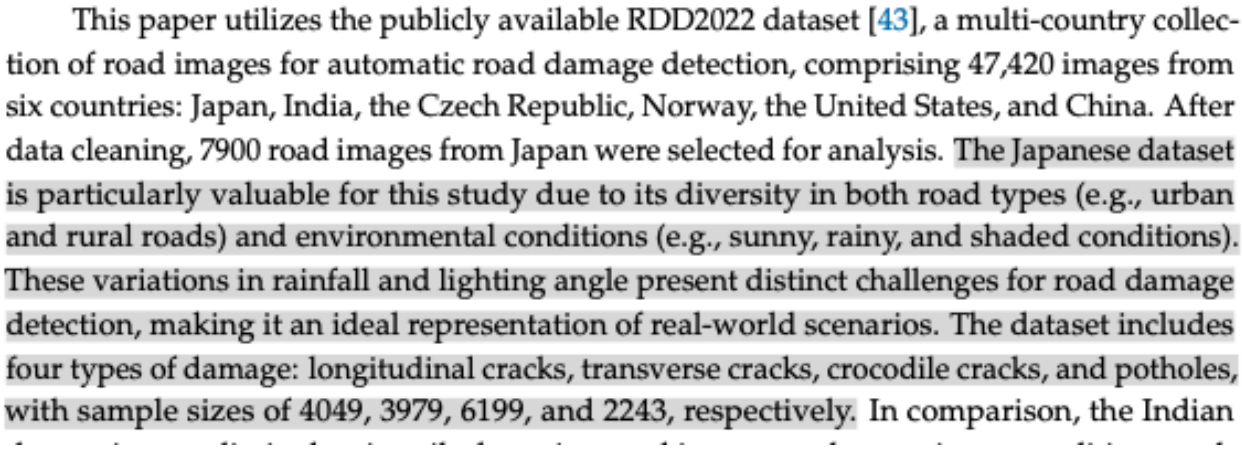
Model	YOLO26s
Core setup	image size 640, batch 16, 100 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, label smoothing 0.1, cache disk, workers 8, AMP true
Augmentation	default
Test	mAP50 0.576, mAP50-95 0.299, precision 0.657, recall 0.524
Test size	2184 images, 5172 instances
Per-class mAP50-95	longitudinal 0.337, transverse 0.263, alligator 0.387, pothole 0.209

Model	YOLO8s
Core setup	image size 640, batch 16, 100 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, label smoothing 0.1, cache disk, workers 8, AMP true
Augmentation	default
Test	mAP50 0.588, mAP50-95 0.302, precision 0.641, recall 0.557
Test size	2184 images, 5172 instances
Per-class mAP50-95	longitudinal 0.333, transverse 0.271, alligator 0.382, pothole 0.221

RDD2022 Japan 4-Class Test Split YOLO26s Experiment `DB-RDD22-JAPAN-4C-V2`

Short name: RDD2022-Japan-4C-TestSplit
Goal: Evaluate YOLO26s on a labeled Japan-only test split under stricter reproduction settings.
Data: Japan only.
Source: `/content/drive/MyDrive/Research/Dataset/Japan_Subset.zip`. Extracted to `/content/Japan_Subset`.
Split: 5530 train images, 1580 validation images, 790 test images. All splits had matching label files.
Classes: 4 classes: longitudinal crack, transverse crack, alligator crack, pothole.

Background images: 0 in test. Not used as a separate class.
Cleaning: YOLO format. `nc=4`. Class IDs found: 0, 1, 2, 3.



This paper utilizes the publicly available RDD2022 dataset [43], a multi-country collection of road images for automatic road damage detection, comprising 47,420 images from six countries: Japan, India, the Czech Republic, Norway, the United States, and China. After data cleaning, 7900 road images from Japan were selected for analysis. The Japanese dataset is particularly valuable for this study due to its diversity in both road types (e.g., urban and rural roads) and environmental conditions (e.g., sunny, rainy, and shaded conditions). These variations in rainfall and lighting angle present distinct challenges for road damage detection, making it an ideal representation of real-world scenarios. The dataset includes four types of damage: longitudinal cracks, transverse cracks, crocodile cracks, and potholes, with sample sizes of 4049, 3979, 6199, and 2243, respectively. In comparison, the Indian

Figure 7: Japan 4-class dataset setup used for the stricter YOLO26s reproduction run.

Table 2. Comparison experiments.

Models	Backbone	mAP	mAP_{50}	mAP_s	mAP_m	mAP_l	Flops/G	Parameters/M
YOLOv5-s	YOLOv5CSPDarknet	18.23	44.02	10.25	15.61	19.93	8.16	7.15
YOLOv7-tiny	E-ELAN	13.51	36.21	6.92	11.36	15.22	6.78	6.15
YOLO X-s	YOLOXCSPDarknet	19.43	46.15	10.86	16.02	21.47	13.54	9.06
YOLOv8-s	YOLOv8CSPDarknet	23.71	52.92	12.73	19.83	27.71	14.27	11.14
YOLOv10-n	CSPDarknet	25.1	53.7	13.78	20.14	29.53	8.25	2.69
Faster R-CNN	ResNet-50	22.01	52.1	9.8	17.9	28.1	101.00	28.29
Cascade R-CNN	ResNet-50	25.03	54.95	12.82	21.16	31.44	162.00	69.16
Ours	CSPDarknet	25.75	55.62	17.66	21.37	29.58	20.28	20.89

Figure 8: Japan 4-class YOLO26s test result comparison.

Run: RUN-JAPAN-4C-001

Model	YOLO26s
Core setup	image size 640, batch 32, 100 epochs
Training	SGD, lr 0.01, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 16, AMP true
Augmentation	Disabled: HSV, translate, scale, fliplr, mosaic, degrees, shear, perspective, flipud, mixup, copy-paste
Test	mAP50 0.390, mAP50-95 0.162, precision 0.495, recall 0.434
Test size	790 images, 1661 instances
Per-class mAP50-95	longitudinal 0.133, transverse 0.112, alligator 0.257, pothole 0.148

Model	YOLO26s
Core setup	image size 640, batch 16, 100 epochs
Training	SGD, lr 0.02, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 8, AMP true
Augmentation	Default
Test	mAP50 0.533, mAP50-95 0.240, precision 0.569, recall 0.543
Test size	790 images, 1661 instances
Per-class mAP50-95	longitudinal 0.222, transverse 0.152, alligator 0.334, pothole 0.253

Model	YOLO8s
Core setup	image size 640, batch 16, 100 epochs
Training	SGD, lr 0.02, momentum 0.937, weight decay 0.0005, warmup 3 epochs, cache disk, workers 8, AMP true
Augmentation	Default
Test	mAP50 0.537, mAP50-95 0.241, precision 0.565, recall 0.523
Test size	790 images, 1661 instances
Per-class mAP50-95	longitudinal 0.225, transverse 0.173, alligator 0.335, pothole 0.232